Comp Maths Sheet 2

(1) It takes n^2 operations to multiply digits together, and another 1 to add them all up. Hence the number of operations is at most n^2+1 so

$$\underline{\text{time complexity} = O(n^2)}$$

(2) a) $$\frac{e_{n+1}}{(e_n)^q} = \frac{x_n - \frac{f(x_n)}{f'(x_n)} - \alpha}{(x_n-\alpha)^q}$$

$$= (e_n)^{-q}\left(e_n - \frac{f(e_n+\alpha)}{f'(e_n+\alpha)}\right)$$

$$= (e_n)^{-q}\left(e_n - \frac{\cancel{f(a)}^{\to 0} + f'(a)e_n + \frac{1}{2}f''(a)e_n^2 + O(e_n^3)}{f'(a) + f''(a)e_n + O(e_n^2)}\right)$$

$$= (e_n)^{2-q}\left(\cancel{1} - \left(\cancel{1} + \frac{f''(a)}{2f'(a)}e_n\right)\right)$$

$$= (e_n)^{2-q}\left(\frac{f''(a)}{2f'(a)}\right) = M$$

Hence $\underline{q = 2}$.

b) If $f'(a) = 0$:

$$\frac{e_{n+1}}{(e_n)^q} = (e_n)^{-q}\left(e_n - \frac{\frac{1}{2}f''(a)e_n^2 + O(e_n^3)}{f''(a)e_n + O(e_n^2)}\right)$$

$$M = (e_n)^{1-q}\left(1 - \frac{1}{2}\right) = \frac{1}{2}(e_n)^{1-q}$$

Hence $q = 1$, converges by $\frac{1}{2}$ each time.

c) If $f'(a) = f''(a) = 0$:

$$\frac{e_{n+1}}{(e_n)^q} = (e_n)^{-q}\left(e_n - \frac{\frac{1}{6}f'''(a)e_n^3 + O(e_n^4)}{\frac{1}{2}f'''(a)e_n^2 + O(e_n^3)}\right)$$

$$= (e_n)^{1-q}\left(1 - \frac{1}{3}\right) = \frac{2}{3}(e_n)^{1-q} = M$$

Hence $q = 1$ (linear), converges by $\frac{2}{3}$ each iteration.

d) $f'(a) \neq 0$, $f''(a) = 0$.

$$\frac{e_{n+1}}{(e_n)^q} = (e_n)^{-q}\left(e_n - \frac{f'(a)e_n + \frac{1}{6}f'''(a)e_n^3 + O(e_n^4)}{f'(a) + \frac{1}{2}f'''(a)e_n^2 + O(e_n^4)}\right)$$

$$= (e_n)^{1-q}\left(1 - \frac{f'''(a)}{3f'(a)}e_n^2 + O(e_n^3)\right)$$

$$\approx (e_n)^{1-q} - \frac{f'''(a)}{3f'(a)}(e_n)^{3-q}.$$

$\therefore$ $q = 3 \Rightarrow$ cubic convergance.